Ujjal Mandal, Ph.D

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Experiences

Oct 2026 – Oct 2026 — **Part Time Lecturer, Department of Physics,** University of Gour Banga, Malda, West Bengal, India.

Sep 2026 – Oct 2026 — **Physics Teacher (Higher Secondary, First and Second Year),** Gurukul – The Vidyapeeth Senior Secondary School (A Unit of Shree Krishna Education and Research Foundation, SKERF), Affiliated with the Assam State School Education Board (ASSEB), Division-II (formerly AHSEC), Phillobai, Tinsukia, Assam – 786160. Teaching Physics following the NCERT curriculum.

Aug 2021– Aug 2026 — **Research Scholar and Teaching Assistantship in Engineering Physics,** Indian Institute of Information Technology Allahabad (An Institute of National Importance, Government of India). Ranked 87th in Engineering in the NIRF 2024 Rankings.

2019 – 2020 — **Physics Teacher (Secondary Level),** Hatmari High School (Government-Sponsored), Affiliated with the West Bengal Board of Secondary Education (WBBSE), Hatimari, Alampur to Eklaxmi Road, West Bengal – 732124.

Education

Aug 2021 – Aug 2026 — **Ph.D. in Computational Biomedical Physics, Ultrasonics and Photoacoustics,** Indian Institute of Information Technology Allahabad (IIIT Allahabad). Degree awarded at the XXI Convocation on 12th September 2026. **Thesis title:** Born Series-Based 2D Helmholtz Equation Solver for Realistic Media in Biomedical Ultrasonics and Photoacoustics.

2020-2022 — **B.Ed.,** West Bengal University of Teachers Training, Education Planning and Administration

2018-2020 — **M.Sc. in Physics,** University of Gour Banga

2015-2018 — **B.Sc. in Physics,** Dum Dum Motijheel College, West Bengal State University

2014 — **Higher Secondary (12th),** West Bengal Council of Higher Secondary Education

2011 — **Secondary Examination (10th),** West Bengal Board of Secondary Education

Entrance Exam Passed

NET-JRF — **Qualified CSIR-UGC NET, Physical Science,** AIR 217, 2020.

JEE (Main) — **Physics,** AIR 1 Lakh, 2014.

WBJEE — **Physics,** State Rank 10000, 2014.

Fellowship

SRF-2023 — **Senior Research Fellowship,** Indian Institute of Information Technology Allahabad

JRF-2021 — **Junior Research Fellowship,** Indian Institute of Information Technology Allahabad

UGC-NFSC — **UGC National Fellowship for Scheduled Caste Students – JRF Fellowship Award,** National Scheduled Castes Finance and Development Corporation

Research Publications

Journal Articles

1. **Ujjal Mandal**, J. Singh, B. T. Cox, and R. K. Saha, "Graphics processing unit accelerated helmholtz equation solver in two dimensions using the traditional born series formulation for linear and nonlinear media in biomedical ultrasound," *The Journal of the Acoustical Society of America*, vol. 159, no. 3, pp. 1948–1962, Mar. 2026, ISSN: 0001-4966. DOI: 10.1121/10.0042817.
2. **Ujjal Mandal**, N. Singh, K. Singh, V. N. Hagone, J. Singh, A. S. Anand, B. T. Cox, and R. K. Saha, "Efficient implementations of a born series for computing photoacoustic field from a collection of erythrocytes," *Photoacoustics*, vol. 43, p. 100 724, 2025, ISSN: 2213-5979. DOI: 10.1016/j.pacs.2025.100724.

Conference Proceedings

1. **Ujjal Mandal**, J. Singh, B. T. Cox, and R. K. Saha, "Pressure field calculation using born series for an array of point sources in inhomogeneous fractional viscoacoustic medium," in *Proceedings of the Annual European Conference on Acoustics and Noise Control Engineering (Forum Acusticum)*, Accepted, 2026. URL: https://www.conforg.fr/bin/v2/htmlview?dir=fa2026&ref=126&lang=1.
2. **Ujjal Mandal**, N. Singh, K. Singh, V. N. Hagone, J. Singh, A. S. Anand, B. T. Cox, and R. K. Saha, "Numerical evidence of multiple scattering of photoacoustic waves in blood," in *Proceedings of the Annual European Conference on Acoustics and Noise Control Engineering (Forum Acusticum)*, Accepted, 2026. URL: https://www.conforg.fr/bin/v2/htmlview?dir=fa2026&ref=124&lang=1.
3. R. K. Saha and **Ujjal Mandal**, "Estimation of optoacoustic spectra for normal and pathological red blood cells using a modified green's function approach," in *Opto-Acoustic Methods and Applications in Biophotonics VI*, Optica Publishing Group, 2023, p. 1 263 109. DOI: 10.1117/12.2670933.
4. **Ujjal Mandal**, J. Singh, and R. K. Saha, "On the born series methods for solving inhomogeneous helmholtz equation in biomedical photoacoustics," in *Opto-Acoustic Methods and Applications in Biophotonics VI*, Optica Publishing Group, 2023, 126310W. DOI: 10.1117/12.2670935.

Skills

Programming	MATLAB, CUDA, C/C++, Python (Basic level)
Software	COMSOL Multiphysics (Basic level), LaTeX, RMarkdown, Beamer, Photoshop, Inkscape

Journal Editorial and Reviewer Role

Editorial Board	**Editorial Board Reviewer,** International Journal of Acoustics and Ultrasonics (IJACUT).
2026 – 2029	**Reviewer,** American Journal of Modern Physics.
Reviewer	**Reviewer,** Journal of Applied Mathematics and Physics (JAMP), Scientific Research Publishing (SCIRP).
	Reviewer, International Journal of Research and Analytical Reviews (IJRAR).
	Reviewer, International Journal of Novel Research and Development (IJNRD).
	Reviewer, TIJER – International Research Journal.
	Reviewer, ScienceOpen.
	Volunteer Reviewer, MDPI Volunteer Reviewer Agreement.
	Reviewer, Science Research.

References

Prof. Ratan K Saha - ratank.saha@iiita.ac.in (Ph.D Supervisior), IIITA

Prof. Jagpreet Singh - jagpreets@iitrpr.ac.in, Department of Computer Science and Engineering, Indian Institute of Technology Ropar (IIT Ropar)